3-amaliy mashg‘ulot.

Determinantning asosiy xossalari. Minor va algebraik to’ldiruvchi tushunchalari. Laplas teoremasi.

Determinantning xossalari:

1. Determinantning satrlarini mos ustunlari bilan almashtirish hatijasida qiymati o‘zgarmaydi.
2. Determinantning biror qatoridagi barcha elementlari nolga teng bo‘lsa, uning qiymati nolga teng bo‘ladi.
3. Determinantning ikkita parrallel qatorining o`rinlarini o‘zaro almashtirish natijasida determinant qiymatining ishorasi qarama-qarshisiga o`zgaradi.
4. Determinantning ikkita parrallel qatori bir xil bo‘lsa, uning qiymati nolga teng bo‘ladi.
5. Agar determinantning biror qatori bir xil ko`paytuvchiga ega bo`lsa, bu ko`paytuvchini determinant belgisidan tashqariga chiqarish mumkin. Demak, determinantni biror songa ko`paytirish uchun uning biror qatori elementlarini shu songa ko`paytirish kifoya.
6. Determinantning ikkita parrallel qatori mos pavishda proporsional bo‘lsa, uning qiymati nolga teng bo‘ladi.

Agar determinantning biror qator elementlari yig`indilardan iborat bo`lsa, u holda bu determinant ikki determinant yig`indisiga teng bo`ladi, bunda birinchi determinantda shu qator birinchi qo`shuluvchilardan, ikkinchisida esa ikkinchi qo`shuluvchilardan tashkil topgan bo`ladi. Masalan,

$$\begin{vmatrix} a_{11}+b_1 & a_{12} & a_{13} \\ a_{21}+b_2 & a_{22} & a_{23} \\ a_{31}+b_3 & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{33} \\ b_3 & a_{32} & a_{33} \end{vmatrix}.$$

7. Agar determinantning biror qatori elementlarini ixtiyoriy songa ko`paytirib, parallel qatori elementlariga mos ravishda qo`shi qiymati o`zgarmaydi.

8. Determinantning qiymati uning biror qatori elementlarini mos algebraik to`ldiruvchilariga ko`paytirilib qo`shilganiga teng.

9. Determinantning biror qatori elementlarini parallel qator mos elementlarining algebraik to`ldiruvchilariga ko'paytmalari yig`indisi nolga teng.

Masalan, $a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} = 0$.

9 - xossa yordamida, (4) formuladan ko`ra umumiyroq bo`lgan, determinantni *biror qatori bo`yicha yoyib hisoblash usuli* hosil bo`ladi. Masalan, uchunchi tartibli determinant uchun

$$\Delta_3 = a_{i1}A_{i1} + a_{i2}A_{i2} + a_{i3}A_{i3}, \qquad (3.1)$$

$$\Delta_3 = a_{1j}A_{1j} + a_{2j}A_{2j} + a_{3j}A_{3j}. \qquad (3.2)$$

Bu yerda (3.1) va (3.1) formulalar mos ravishda determinantning ixtiyoriy $i-$ *satri* va $j-$ *ustuni bo`yicha yoyilmasi* deyiladi.

Determinantning a_{ij} elementining M_{ij} *minori* deb, uning $i-$ satri va $j-$ ustunini o`chirishdan hosil bo`lgan determinantga aytiladi.

Masalan, uchunchi tartibli determinant uchun

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, \qquad M_{31} = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}.$$

Determinantning a_{ij} elementining A_{ij} *algebraik to`ldiruvchisi* deb, $A_{ij} = (-1)^{i+j} M_{ij}$ tenglik bilan aniqlanadigan songa aytiladi.

Masalan, uchunchi tartibli determinant uchun

$$A_{21} = (-1)^{2+1} M_{21} = -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}, \qquad A_{22} = (-1)^{2+2} M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}.$$

1-misol. $\begin{vmatrix} 1 & 1 & 4 \\ -1 & 2 & 3 \\ -3 & 2 & 5 \end{vmatrix}$ determinantning M_{23} minorini hisoblang.

► Determinantning $2-$ satri va $3-$ ustunini o`chiramiz:

$$\begin{vmatrix} 1 & 1 & 4 \\ -1 & 2 & 3 \\ -3 & 2 & 5 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ -3 & 2 \end{vmatrix} = 1\cdot 2 - 1\cdot(-3) = 2+3 = 5.$$ Demak, $M_{23} = 5$. ◄

2-misol. $\begin{vmatrix} 2 & 1 & -4 \\ 1 & 3 & 5 \\ 3 & 2 & -1 \end{vmatrix}$ determinantning A_{32} va A_{13} algebraik to`ldiruvchilarini hisoblang.

► $A_{32} = (-1)^{3+2} M_{32}$, ya'ni $A_{32} = -M_{32}$ bo`lgani uchun, determinantning $3-$ satri va $2-$ ustunini o`chiramiz:

$$A_{32} = -\begin{vmatrix} 2 & 1 & -4 \\ 1 & 3 & 5 \\ 3 & 2 & -1 \end{vmatrix} = -\begin{vmatrix} 2 & -4 \\ 1 & 5 \end{vmatrix} = -(2\cdot 5 - (-4)\cdot 1) = -14 \cdot$$

$A_{13} = (-1)^{1+3} M_{32}$ yoki $A_{13} = M_{13}$ bo`lgani uchun, determinantning $1-$satri va $3-$ ustunini o`chirib hisoblaymiz.

$$A_{13} = \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} = 1\cdot 2 - 3\cdot 3 = -7.$$

Demak, $A_{32} = -14$, $A_{13} = -7$. ◄

(3.1) formulani algebraik to`lduruvchilar yordamida quyidagicha ifodalaymiz:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} \qquad (3.3)$$

3-misol. $\begin{vmatrix} 2 & 5 & -4 \\ 1 & -3 & 5 \\ 3 & 2 & -1 \end{vmatrix}$ determinantni hisoblang.

► (3.3) formulani qo`llaymiz, buning uchun avval A_{11} hisoblaymiz:

$$A_{11}=\begin{vmatrix}-3 & 5\\ 2 & -1\end{vmatrix}=3-10=-7,\ A_{12}=-\begin{vmatrix}1 & 5\\ 3 & -1\end{vmatrix}=-(-1-15)=16,$$

$$A_{13}=\begin{vmatrix}1 & -3\\ 3 & 2\end{vmatrix}=2-(-9)=11.$$

$$\begin{vmatrix}2 & 5 & -4\\ 1 & -3 & 5\\ 3 & 2 & -1\end{vmatrix}=2\cdot(-7)+5\cdot 16-4\cdot 11=-14+80-44=22.\blacktriangleleft$$

4-misol. $\begin{vmatrix}2 & 5 & -4\\ 1 & -3 & 5\\ 3 & 2 & -1\end{vmatrix}$ determinantni hisoblang.

► (3.3) formulani qo`llaymiz, buning uchun avval A_{11}, A_{12} va A_{13} larni hisoblaymiz:

$$A_{11}=\begin{vmatrix}-3 & 5\\ 2 & -1\end{vmatrix}=3-10=-7,\ A_{12}=-\begin{vmatrix}1 & 5\\ 3 & -1\end{vmatrix}=-(-1-15)=16,$$

$$A_{13}=\begin{vmatrix}1 & -3\\ 3 & 2\end{vmatrix}=2-(-9)=11.$$

$$\begin{vmatrix}2 & 5 & -4\\ 1 & -3 & 5\\ 3 & 2 & -1\end{vmatrix}=2\cdot(-7)+5\cdot 16-4\cdot 11=-14+80-44=22.\blacktriangleleft$$

5-misol. $\begin{vmatrix}4 & -2 & 0\\ 3 & 5 & 6\\ -3 & 4 & 0\end{vmatrix}$ determinantni biror qatori bo`yicha yoyib hisoblang.

► Determinantni eng ko`p nol element qatnashgan qatorini aniqlaymiz. Bu yerda uchunchi ustunda eng ko`p nol element bo`lgani uchun, (3.2) formulani qo`llaymiz: $\Delta_3=a_{13}A_{13}+a_{23}A_{23}+a_{33}A_{33}=6A_{23}$, chunki $a_{13}=a_{33}=0$.

$$\begin{vmatrix}4 & -2 & 0\\ 3 & 5 & -6\\ -3 & 4 & 0\end{vmatrix}=-6\cdot(-1)^{2+3}\begin{vmatrix}4 & -2\\ -3 & 4\end{vmatrix}=6\cdot(16-6)=60.\blacktriangleleft$$

6-misol. Byerilgan Δ determinant uchun a_{12}, a_{32} elyemyentlarning minorlari va algyebraik to'ldiruvchilarni toping. Δ determinantni: a) birinchi satr elyemyentlari bo'yicha yoyib; b) ikkinchi ustun elyemyentlari bo'yicha yoyib; v) birinchi satr elyemyentlarini nolga aylantirib, hisoblang.

$$\Delta = \begin{vmatrix} -3 & 2 & 1 & 0 \\ 2 & -2 & 1 & 4 \\ 4 & 0 & -1 & 2 \\ 3 & 1 & -1 & 4 \end{vmatrix}$$

Yechilishi. Quyidagilarni topamiz:

$$M_{12} = \begin{vmatrix} 2 & 1 & 4 \\ 4 & -1 & 2 \\ 3 & -1 & 4 \end{vmatrix} = -8 - 16 + 6 + 12 + 4 - 16 = -18.$$

$$M_{32} = \begin{vmatrix} -3 & 1 & 0 \\ 2 & 1 & 4 \\ 3 & -1 & 4 \end{vmatrix} = -12 + 12 - 12 - 8 = -20.$$

a_{12}, a_{32} elyemyentlarning algyebraik to'ldiruvchilari mos ravishda quyidagilarga tyeng:

$$A_{12} = (-1)^{1+2} M_{12} = -(-18) = 18.$$

$$A_{32} = (-1)^{3+2} M_{32} = -(-20) = 20.$$

a) Δ determinantni birinchi satr elyemyentlari bo'yicha yoyib hisoblaymiz:

$$\Delta = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + a_{14}A_{14} =$$

$$= -3\begin{vmatrix} -2 & 1 & 4 \\ 0 & -1 & 2 \\ 1 & -1 & 4 \end{vmatrix} - 2\begin{vmatrix} 2 & 1 & 4 \\ 4 & -1 & 2 \\ 3 & -1 & 4 \end{vmatrix} + 1\begin{vmatrix} 2 & -2 & 4 \\ 4 & 0 & 2 \\ 3 & 1 & 4 \end{vmatrix} =$$

$$= -3(8+2+4-4) - 2(-8 - 16 + 6 + 12 + 4 - 16) + (16 - 12 - 4$$

b) Δ determinantni ikkinchi ustun elyemyentlari bo'yicha yoyib hiso

$$\Delta = -2\begin{vmatrix} 2 & 1 & 4 \\ 4 & -1 & 2 \\ 3 & -1 & 4 \end{vmatrix} - 2\begin{vmatrix} -3 & 1 & 0 \\ 4 & -1 & 2 \\ 3 & -1 & 4 \end{vmatrix} + 1\begin{vmatrix} -3 & 1 & 0 \\ 2 & 1 & 4 \\ 4 & -1 & 2 \end{vmatrix} =$$

$$= -2(-8 + 6 - 16 + 12 + 4 - 16) - 2(12 + 6 - 6 - 16) + (-6 + 16 - 12 - 4) = 38;$$

d) Δ determinantni birinchi satr elyemyentlarini nolga aylantirib hisoblaymiz. Dyetyerminantning uchinchi ustunini 3 ga ko'paytiramiz va birinchi ustunga qo'shamiz, so'ngra uchinchi ustunini -2 ga ko'paytiramiz va ikkinchi ustunga qo'shamiz. U holda birinchi satrning bitta elyemyentidan boshqa barcha elyemyentlari nollardan iborat bo'ladi. Hosil bo'lgan determinantni birinchi satr elyemyentlari bo'yicha yoyib hisoblaymiz:

$$\Delta = \begin{vmatrix} -3 & 2 & 1 & 0 \\ 2 & -2 & 1 & 4 \\ 4 & 0 & -1 & 4 \\ 3 & 1 & -1 & 4 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 & 0 \\ 5 & -4 & 1 & 4 \\ 1 & 2 & -1 & 2 \\ 0 & 3 & -1 & 4 \end{vmatrix} = \begin{vmatrix} 5 & -4 & 4 \\ 1 & 2 & 2 \\ 0 & 3 & 4 \end{vmatrix} = \begin{vmatrix} 0 & -14 & -6 \\ 1 & 2 & 2 \\ 0 & 3 & 4 \end{vmatrix} =$$

$$= -(-56 + 18) = 38.$$

Yuqorida uchinchi tartibli determinantning birinchi ustunida nollarni hosil qilib hisobladik. ■

1. Yuqori tartibli determinantni, asosiy xossalaridan foydalanib, biror qatorining bitta elementidan boshqa barcha elementlari nolga aylantirilib, so`ng 9-xossa yordamida *tartibini pasaytirib hisoblash* mumkin.

7-misol. $\Delta = \begin{vmatrix} 2 & -4 & 1 & 5 \\ 1 & -3 & 2 & 5 \\ 2 & 2 & 0 & -3 \\ 3 & -1 & 1 & 2 \end{vmatrix}$ determinantni hisoblang.

► Determinantning birinchi satrini -2 va -1 ga ko`paytirib, mos ravishda ikkinchi va to`rtinchi satriga qo`shamiz:

$$\Delta = \begin{vmatrix} 2 & -4 & 1 & 5 \\ -3 & 5 & 0 & -5 \\ 2 & 2 & 0 & -3 \\ 1 & 3 & 0 & -3 \end{vmatrix}$$

$3-$ustun bo`yicha yoyib(9-xossa), ya'ni $\Delta = a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33} + a_{43}A_{43}$ va $a_{23} = a_{33} = a_{43} = 0$ ekanini e'tiborga olib, tartibini pasaytiramiz. Hosil bo`lgan uchunchi tartibli determinantni esa uchburchak usulida yechamiz.

$$\Delta = 1 \cdot (-1)^{1+3} \begin{vmatrix} -3 & 5 & -5 \\ 2 & 2 & -3 \\ 1 & 3 & -3 \end{vmatrix} = 18 - 15 - 30 + 10 + 30 - 27 = -14 . \blacktriangleleft$$

2. Bosh diagonalidan yuqorisidagi yoki pastidagi barcha elementlari nollardan iborat bo`lgan determinant *uchburchak shaklidagi determinant* deyiladi. Bunday determinantning qiymati bosh diagonali elementlari ko`paytmasiga teng. Har qanday determinantni *uchburchak shakliga keltirib hisoblash* mumkin.

8-misol. $\Delta = \begin{vmatrix} 2 & -4 & 1 & 5 \\ 1 & -3 & 2 & 5 \\ 2 & 2 & 0 & -3 \\ 3 & -1 & 1 & 2 \end{vmatrix}$ determinantni uchburchak shakliga keltirib hisoblang.

► Determinantning a_{11} elementini 1 ga aylantirish uchun birinchi va ikkinchi satrlarining o`rinlarini almashtiramiz. Hosil bo`lgan birinchi satrni -2, -2 va -3 ga ko`paytirib, mos ravishda ikkinchi , uchunchi va to`rtinchi satrlarga qo`shamiz.

$$\Delta = -\begin{vmatrix} 1 & -3 & 2 & 5 \\ 2 & -4 & 1 & 5 \\ 2 & 2 & 0 & -3 \\ 3 & -1 & 1 & 2 \end{vmatrix} = -\begin{vmatrix} 1 & -3 & 2 & 5 \\ 0 & 2 & -3 & -5 \\ 0 & 8 & -4 & -13 \\ 0 & 8 & -5 & -13 \end{vmatrix}$$

Uchinchi satrini -1 ga ko`paytirib to`rtinchi satrlarga qo`shamiz:

$$\Delta = -\begin{vmatrix} 1 & -3 & 2 & 5 \\ 0 & 2 & -3 & -5 \\ 0 & 8 & -4 & -13 \\ 0 & 0 & -1 & 0 \end{vmatrix}$$

Ikkinchi satrini -4 ga ko`paytirib uchinchi satriga qo`shamiz. So`ng uchinchi va to`rtinchi satrlar o`rinlarini almashtiramiz:

$$\Delta = -\begin{vmatrix} 1 & -3 & 2 & 5 \\ 0 & 2 & -3 & -5 \\ 0 & 0 & 8 & 7 \\ 0 & 0 & -1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -3 & 2 & 5 \\ 0 & 2 & -3 & -5 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 8 & 7 \end{vmatrix}$$

Uchinchi satrini 8 ga ko`paytirib to`rtinchi satriga qo`shamiz :

$$\Delta = \begin{vmatrix} 1 & -3 & 2 & 5 \\ 0 & 2 & -3 & -5 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 7 \end{vmatrix} = 1 \cdot 2 \cdot (-1) \cdot 7 = -14. \blacktriangleleft$$

9-misol. Determinantni hisoblang: $\begin{vmatrix} 2384 & 1384 \\ 2385 & 1385 \end{vmatrix}$.

Yechish. Determinant xossasiga ko'ra uning birinchi satrini -1ga ko'paytirib ikkinchi satrga mos ravishda qo'shsak,

$$\begin{vmatrix} 2384 & 1384 \\ 2385 & 1385 \end{vmatrix} = \begin{vmatrix} 2384 & 1384 \\ 1 & 1 \end{vmatrix} = 2384 - 1384 - 1000.$$

Auditoriya topshiriqlari

1. Tenglikni determinant xossalaridan foydalanib tekshiring.

$$a)\ \begin{vmatrix} 1 & 3 & 1 \\ 7 & 0 & 7 \\ 6 & 4 & 6 \end{vmatrix} = 0 \qquad b)\ \begin{vmatrix} 5 & 6 & 1 \\ 5 & 2 & 3 \\ 11 & 5 & 0 \end{vmatrix} = \begin{vmatrix} 3 & 6 & 1 \\ 4 & 2 & 3 \\ 8 & 5 & 0 \end{vmatrix} = \begin{vmatrix} 2 & 6 & 1 \\ 1 & 2 & 3 \\ 3 & 5 & 0 \end{vmatrix}$$

$$c)\ \begin{vmatrix} 4 & 6 & 0 \\ 2 & 7 & 3 \\ 1 & 5 & 1 \end{vmatrix} = \begin{vmatrix} 4 & 6 & 0 \\ 2+2*3 & 7 & 3 \\ 1+2*1 & 5 & 1 \end{vmatrix} \qquad d)\ \begin{vmatrix} 1 & 3 & 4 \\ 7 & 3 & 1 \\ 2 & 9 & 4 \end{vmatrix} = -\begin{vmatrix} 4 & 3 & \\ 1 & 3 & \\ 4 & 9 & \end{vmatrix}$$

2. Determinantni yoymasdan isbotlang.

$$\begin{vmatrix} \beta b_1+\gamma c_1 & b_1 & c_1 \\ \beta b_2+\gamma c_2 & b_2 & c_2 \\ \beta b_3+\gamma c_3 & b_3 & c_3 \end{vmatrix}=0$$

3. Berilgan ikkinchi tartibli deteminantlarni hisoblang.

a) $\begin{vmatrix} 5 & 3 \\ 7 & -4 \end{vmatrix}$; b) $\begin{vmatrix} 4 & -7 \\ -2 & -3 \end{vmatrix}$; d) $\begin{vmatrix} tgx & -1 \\ 1 & tgx \end{vmatrix}$.

4. Tenglamani yeching.

a) $\begin{vmatrix} x+3 & 2 \\ 7 & x-2 \end{vmatrix}=0$; b) $\begin{vmatrix} \sin 2x & -\cos 2x \\ \sin 3x & \cos 3x \end{vmatrix}=0$.

5. Berilgan uchinchi tartibli deteminantlarni hisoblang.

a) $\begin{vmatrix} 1 & -2 & 4 \\ -3 & 5 & 5 \\ 2 & -1 & 3 \end{vmatrix}$; b) $\begin{vmatrix} 10 & -2 & 4 \\ -15 & 3 & 6 \\ 20 & -1 & 5 \end{vmatrix}$; d) $\begin{vmatrix} 1 & 2 & 5 \\ 5 & -3 & 7 \\ 4 & 6 & 5 \end{vmatrix}$.

6. Berilgan uchinchi tartibli deteminantlarni satr yoki ustun bo`yicha yoyib hisoblang.

a) $\begin{vmatrix} 5 & 0 & 6 \\ 4 & 0 & 5 \\ 2 & 4 & 3 \end{vmatrix}$; b) $\begin{vmatrix} 2 & 2 & -1 \\ 7 & 0 & 3 \\ 3 & 4 & 0 \end{vmatrix}$, d) $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{vmatrix}$.

7. Berilgan deteminantlarni uchburchak shakliga keltirib hisoblang.

a) $\begin{vmatrix} 2 & -3 & 2 & 4 \\ -3 & 2 & 2 & 5 \\ 1 & 5 & -3 & 0 \\ 0 & -1 & 1 & 2 \end{vmatrix}$; b) $\begin{vmatrix} 4 & 0 & -3 & 5 \\ 3 & 2 & -2 & 1 \\ 1 & 3 & 1 & 0 \\ 5 & 6 & 2 & -1 \end{vmatrix}$.

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